Early Vision
(I)

Introduction to Computational and Biological Vision

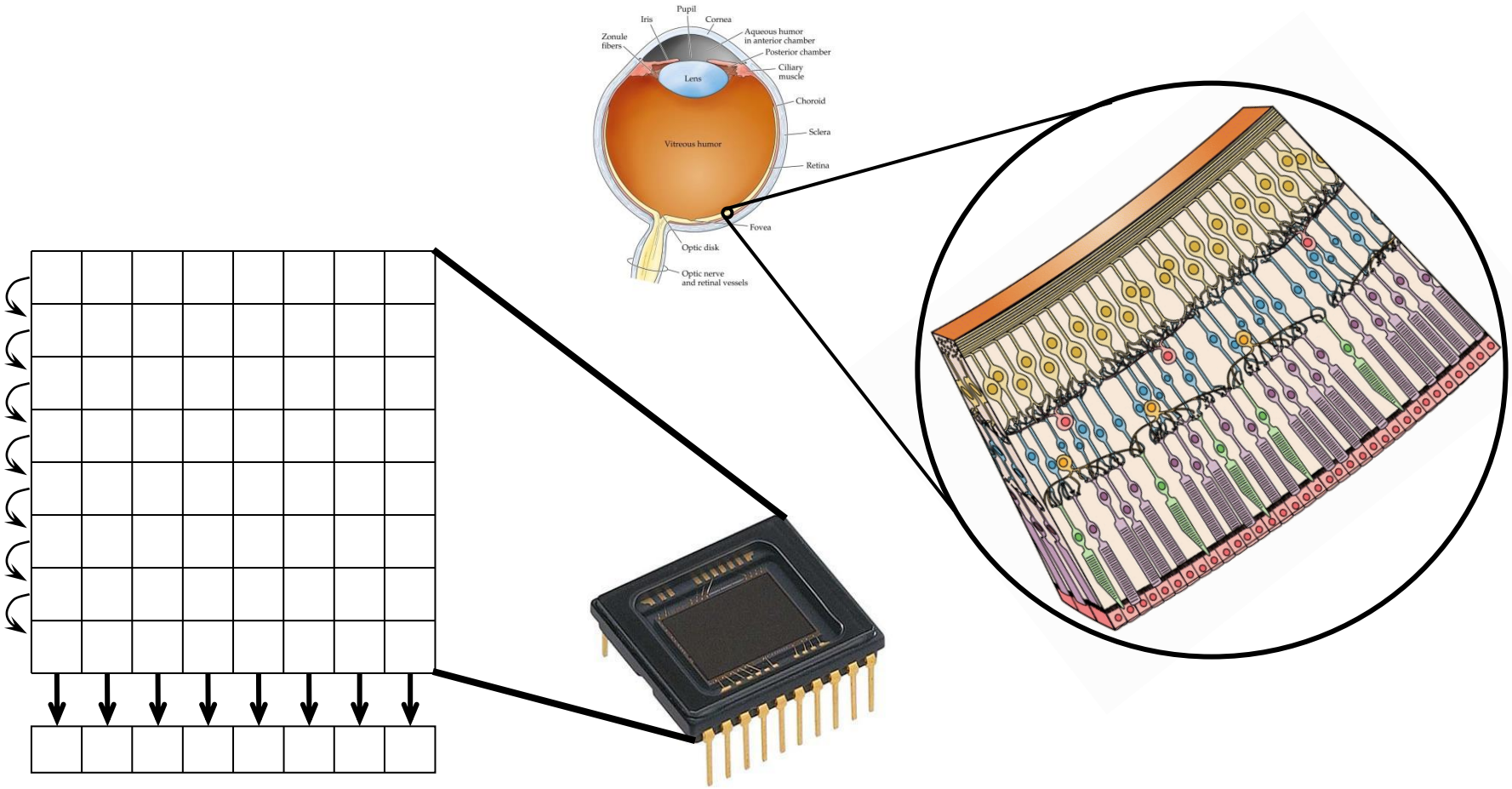
CS 202-1-5261

Computer Science Department, BGU

Ohad Ben-Shahar

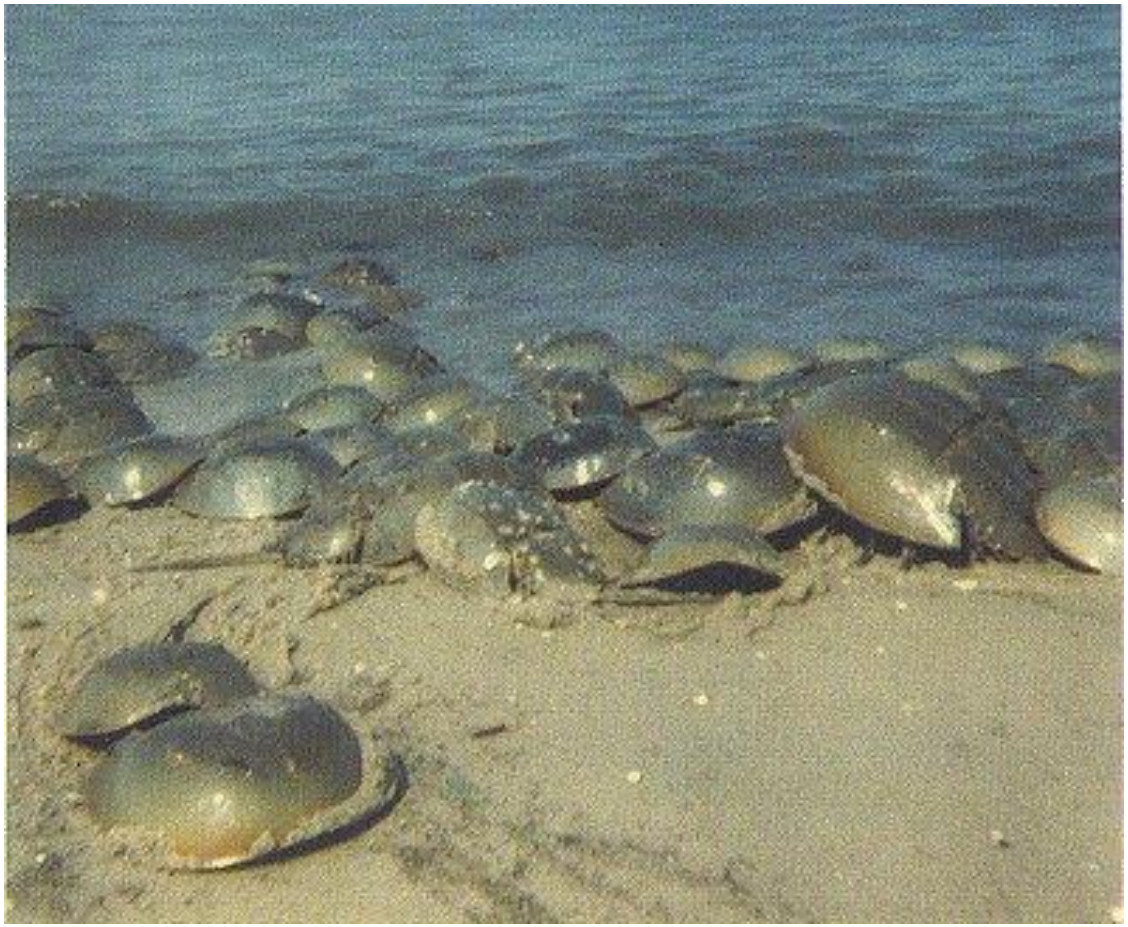
Processing at the sensor level

How different are these designs?



Processing at the sensor level

Limulus Polyphemus (Horseshoe crab)



Processing at the sensor level

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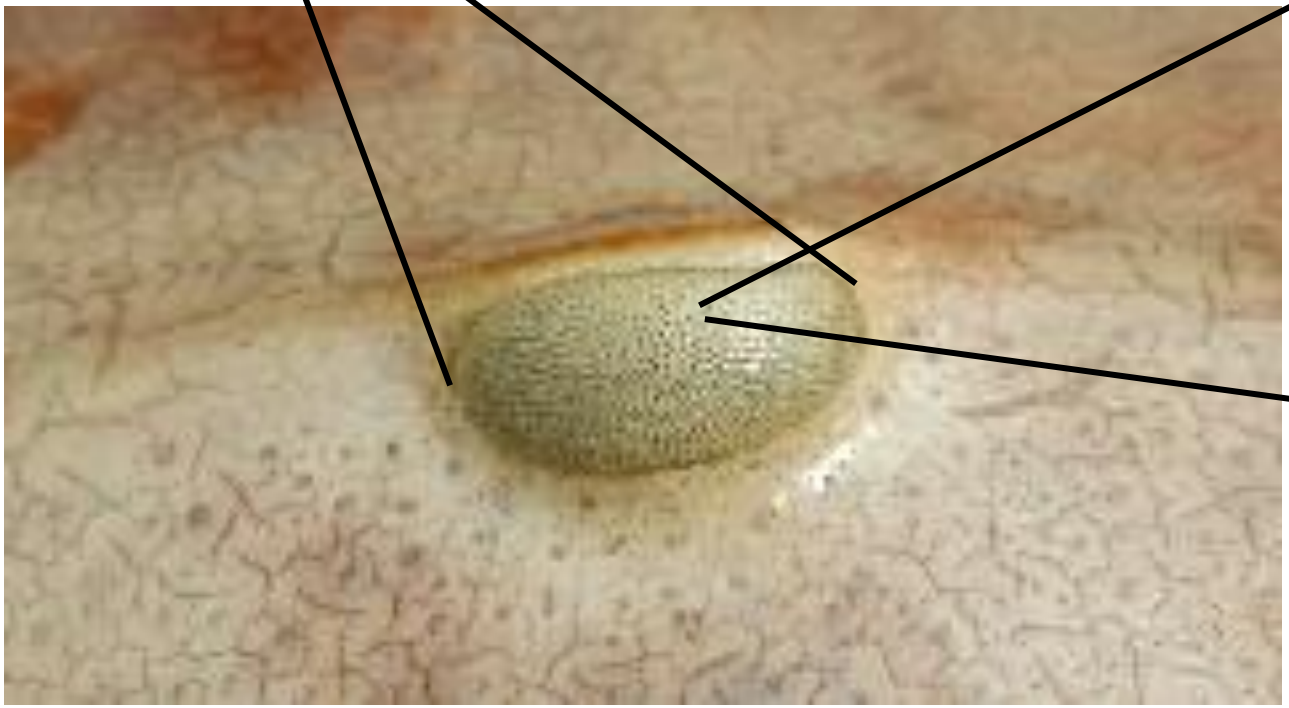

Processing at the sensor level

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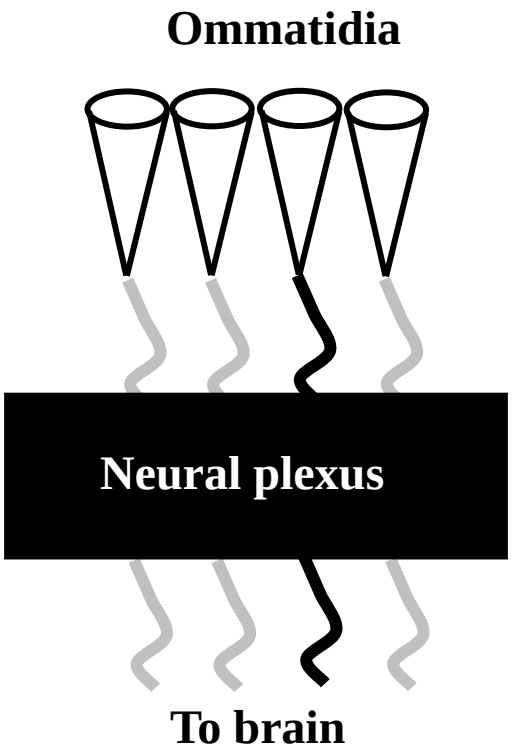
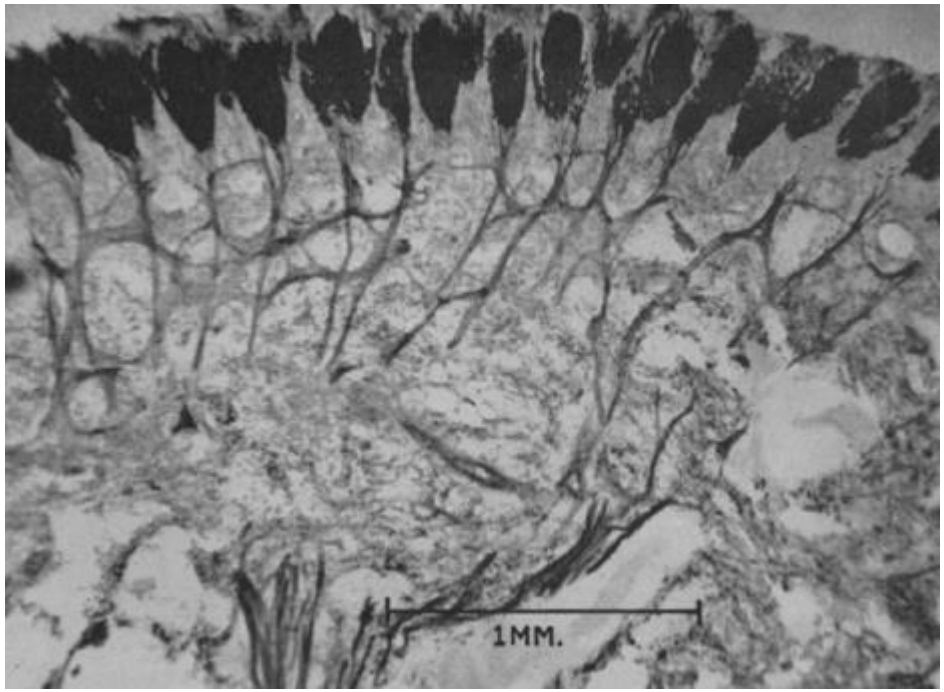
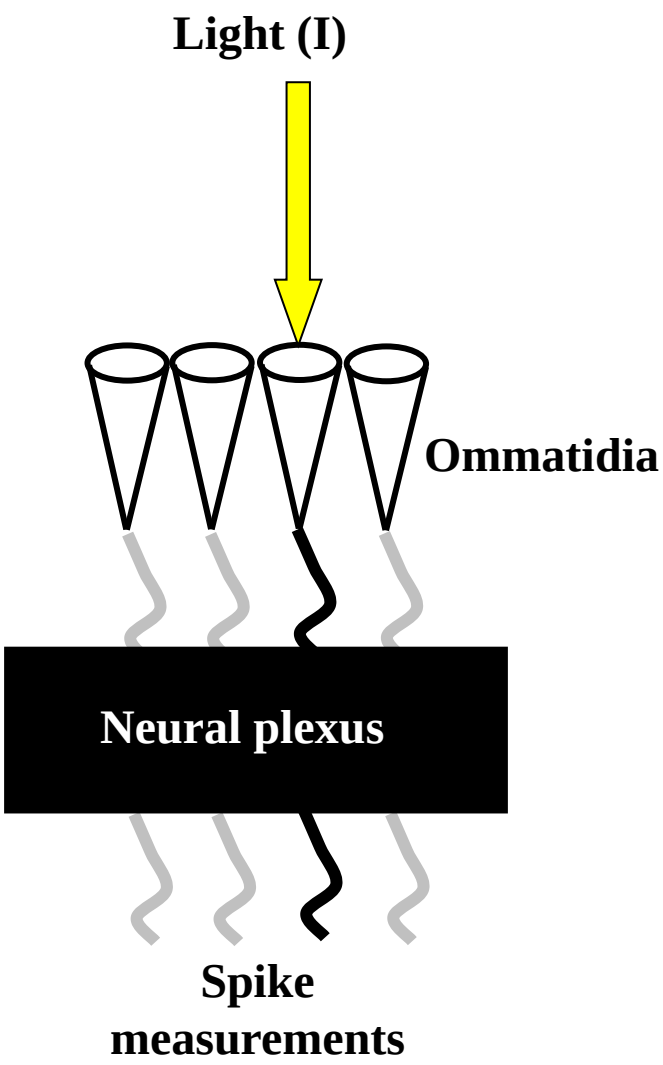
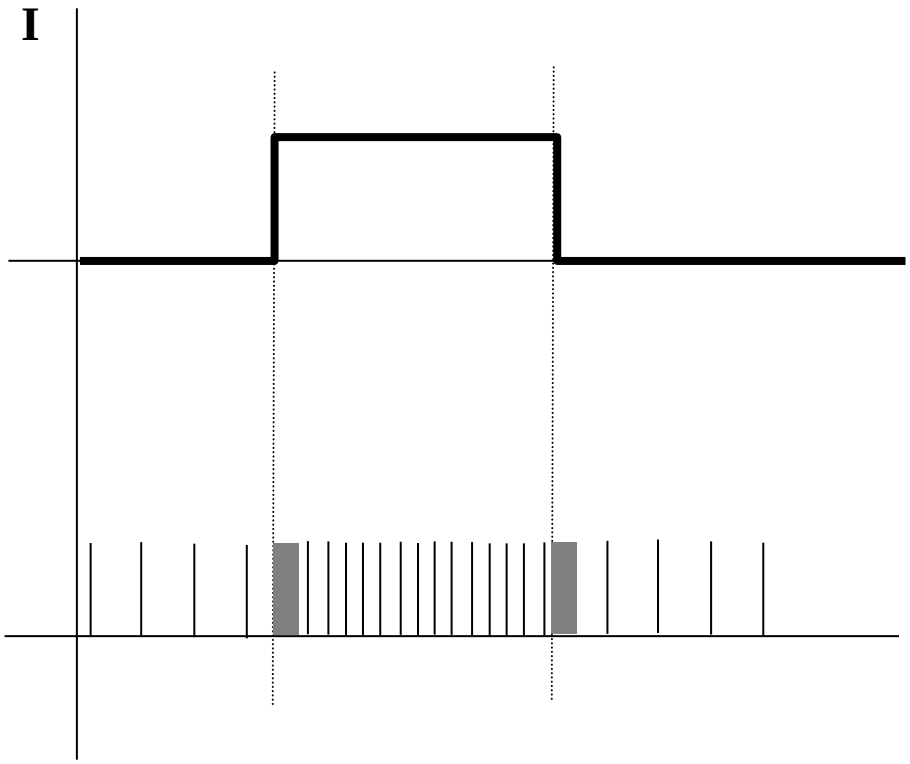




Processing at the sensor level



Haldan Hartline's experiments



Processing at the sensor level

(Classical) receptive field of a cell

A region of the retina (visual field)

that must be stimulated directly

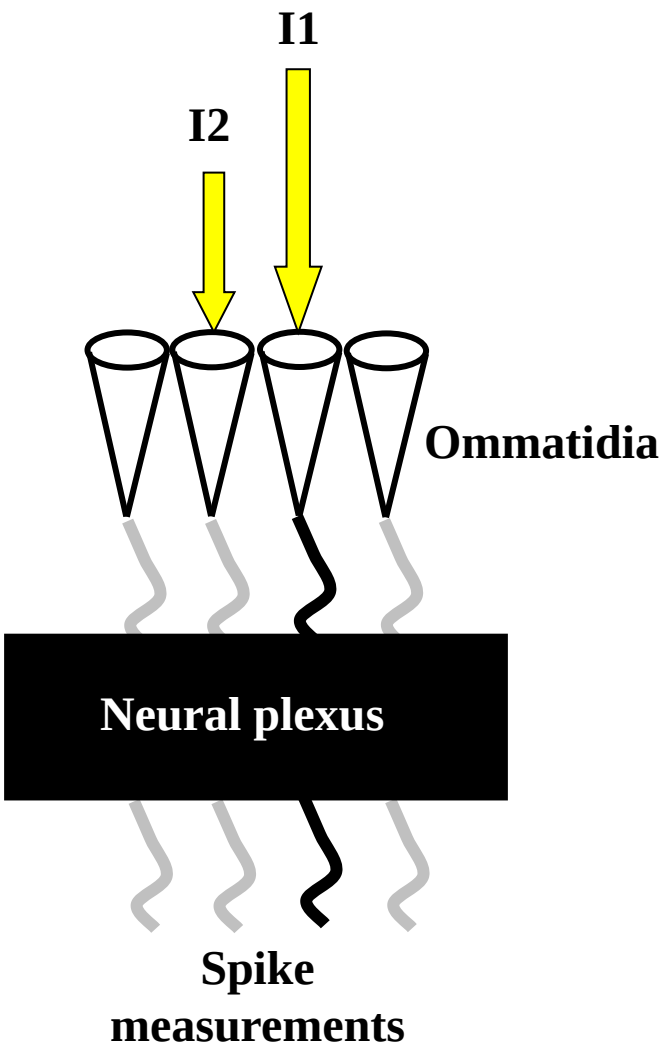
in order to obtain a response in that cell

[Sherrington 1906]

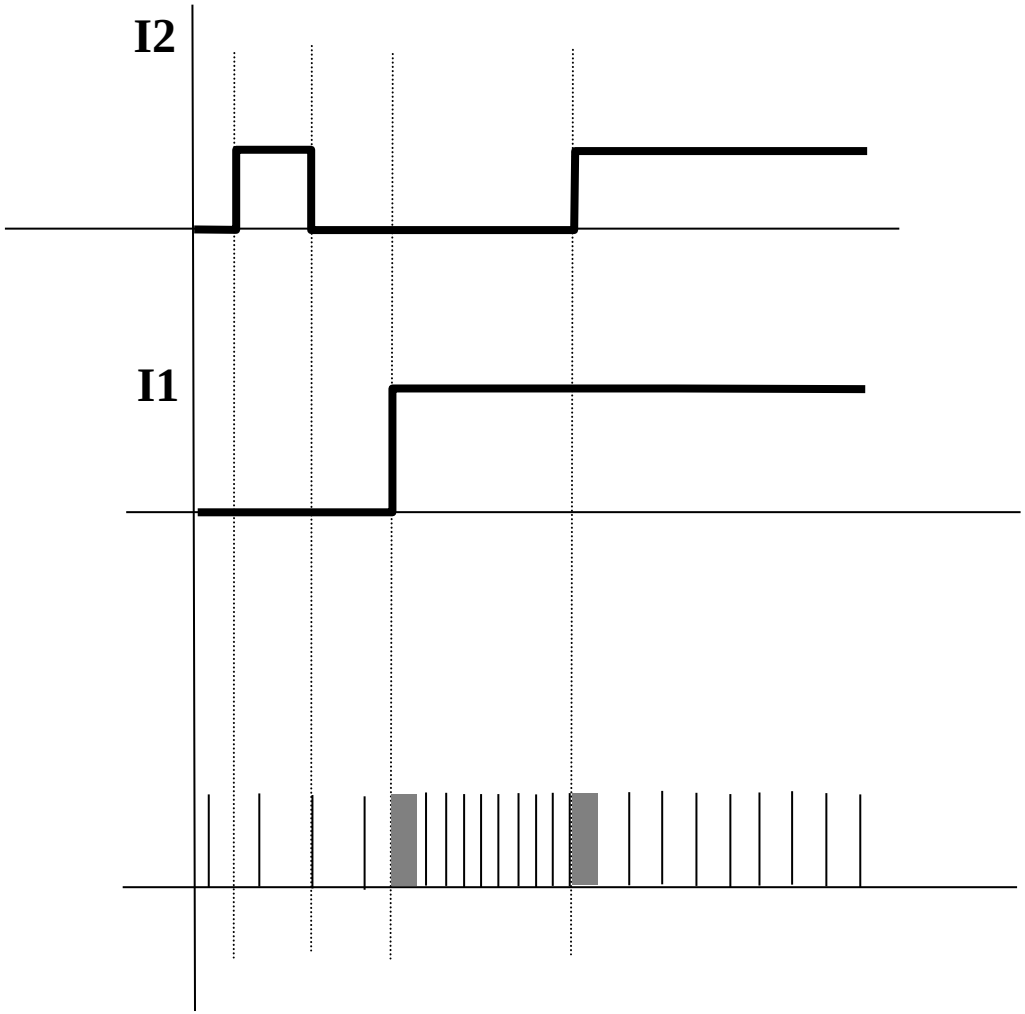
[Hartline 1938]

[Kuffler 1953]

Processing at the sensor level

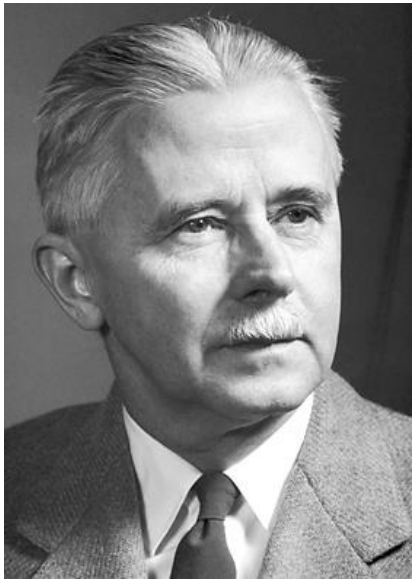


Hartline's *lateral inhibition*



Processing at the sensor level

Lateral inhibition in the Limulus

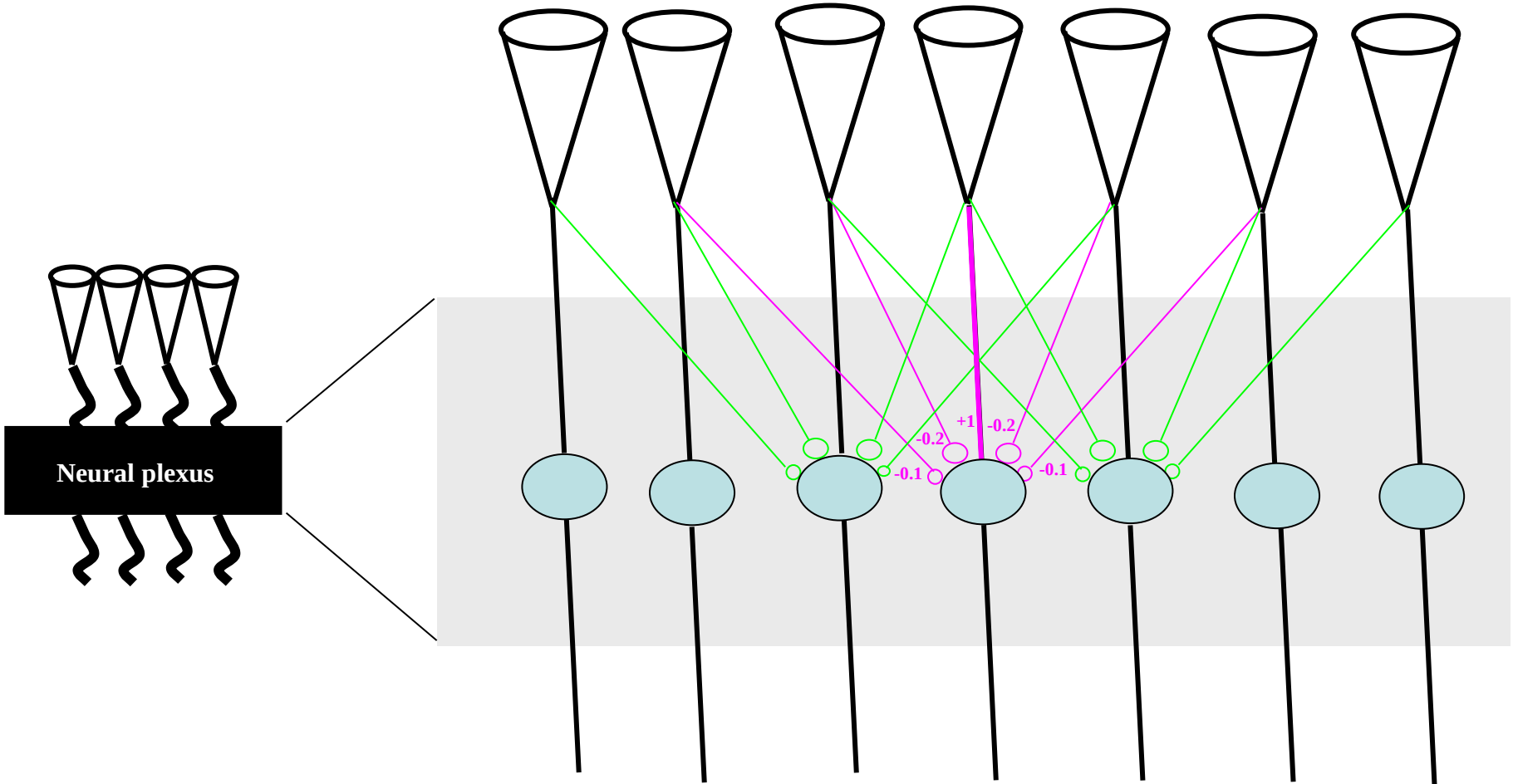


- **Nearby receptors mutually inhibit each other.**
- **Inhibition increases with the intensity of the inhibiting illumination.**
- **Inhibition increases with the area of the inhibiting illumination.**
- **Inhibition diminishes with the distance of the inhibiting illumination.**
- **Inhibition is independent of excitation level**

[Hartline, Wagner, and Ratliff 1956]

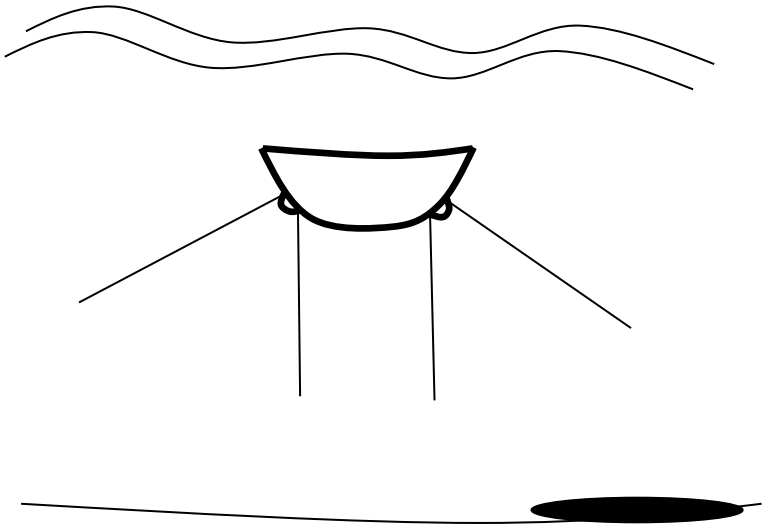
Processing at the sensor level

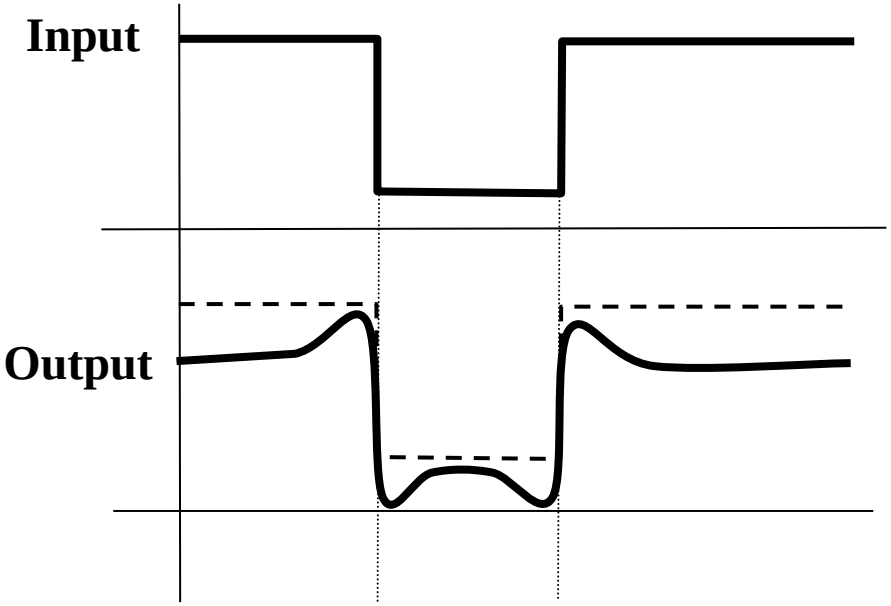
Lateral inhibition in the Limulus – HOW?



Processing at the sensor level

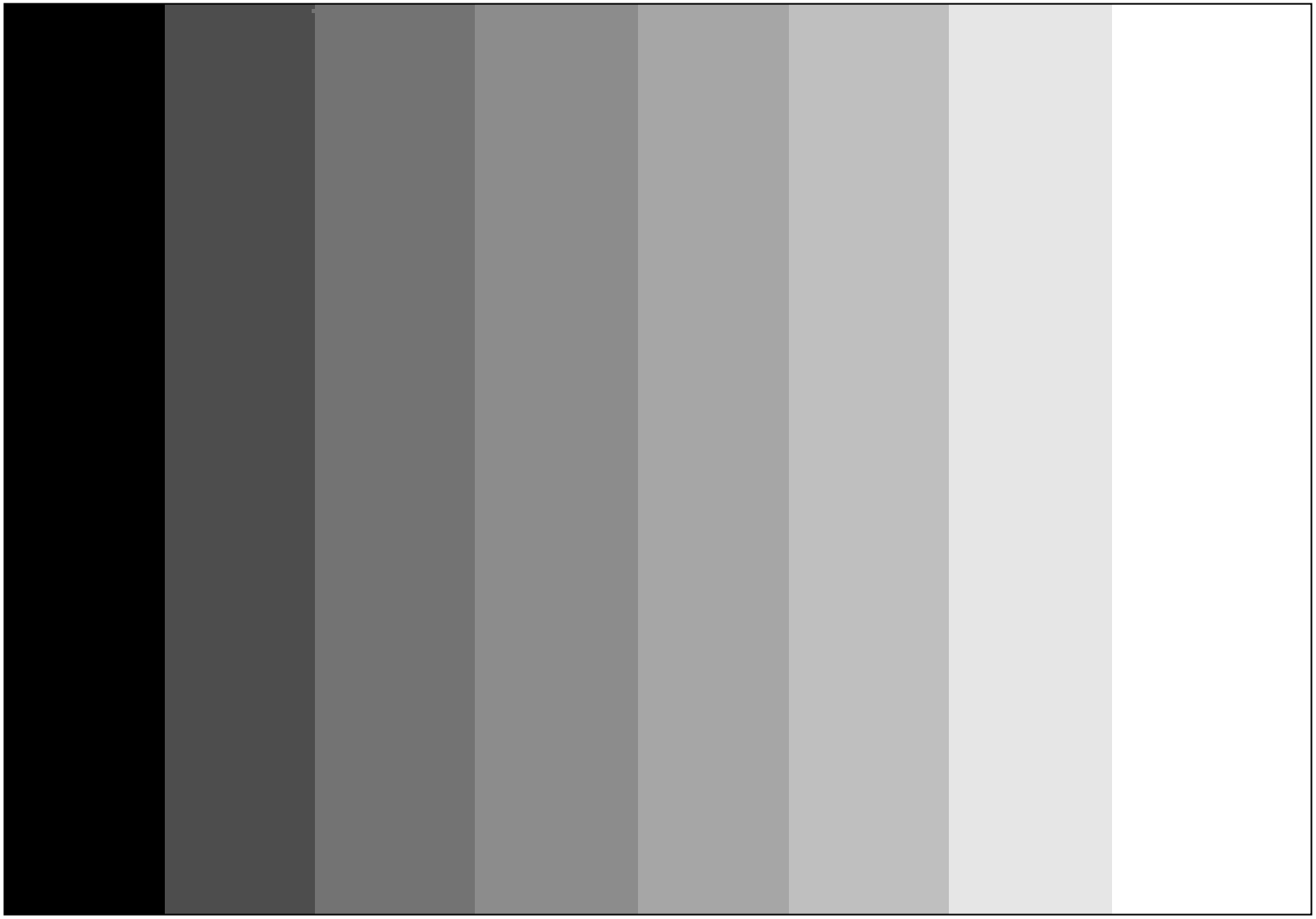
Why lateral inhibition?





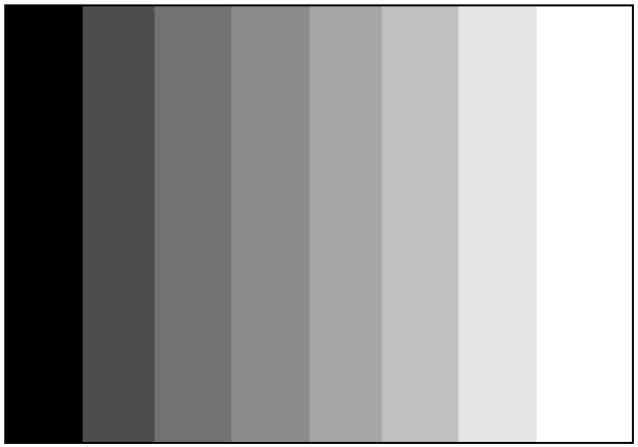
Contrast and boundary enhancement!!

Processing at the sensor level



Processing at the sensor level

Mach Bands



Ernst Mach (1838-1916)

Processing at the sensor level

What can be computed with lateral inhibition?

What can be computed with lateral *interactions*?

What cannot be computed?

What is an appropriate abstraction and formalism?